


Figure 5.22 Multistage transistor amplifiers

arrangement, i.e., output of the first-stage is connected to the input of the second-stage through a suitable coupling device, and so on. Figure 5.22 shows the block diagram of the multistage transistor amplifier.

## Coupling of Amplifiers

| Name of Coupling | Name of Multistage Amplifier |
|---|---|
| RC coupling | RC coupled amplifier |
| Transformer coupling | Transformer coupled amplifier |
| Direct coupling | Direct coupled amplifier |

### 5.10.1 RC Coupled Transistor Amplifier

This is the most popular type of coupling because it is cheap and provides excellent audio fidelity over a wide range of frequency. It is usually employed for voltage amplification.

Figure 5.23 shows two stages of an RC coupled amplifier. A coupling capacitor $C_C$ is used to connect the output of first-stage to the base (i.e. input) of the second-stage and so on. As the coupling from one stage to next is achieved by a coupling capacitor followed by a connection to a shunt resistor, therefore, such amplifiers are called resistance-capacitance coupled amplifiers.

**Operation:** When AC signal is applied to the base of the first transistors, it appears in the amplified form across its collector load $R_C$. Then, the amplified signal is given to the base of next stage through coupling capacitor $C_C$. The second stage does further amplification of the signal. In this way, the cascaded (one after another) stages amplify the signal and the overall gain is considerably increased.

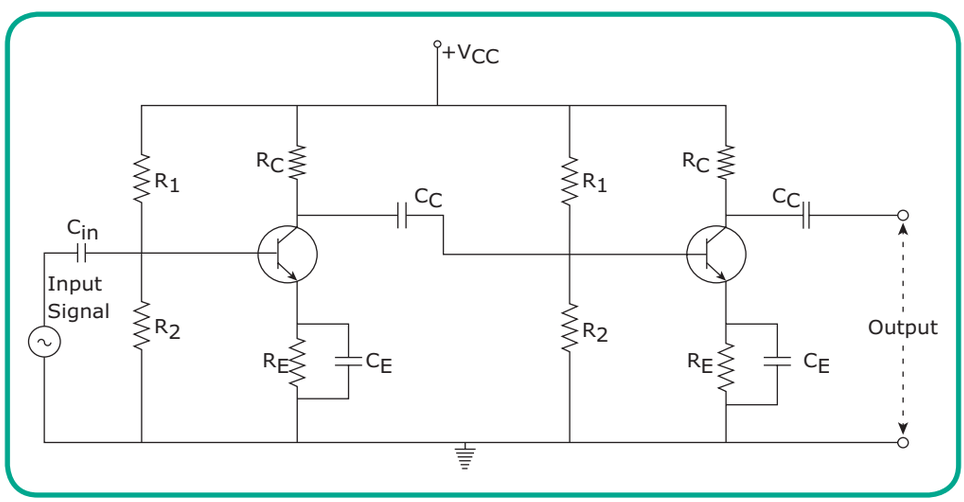



Figure 5.23 Stages of an RC coupled amplifier



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